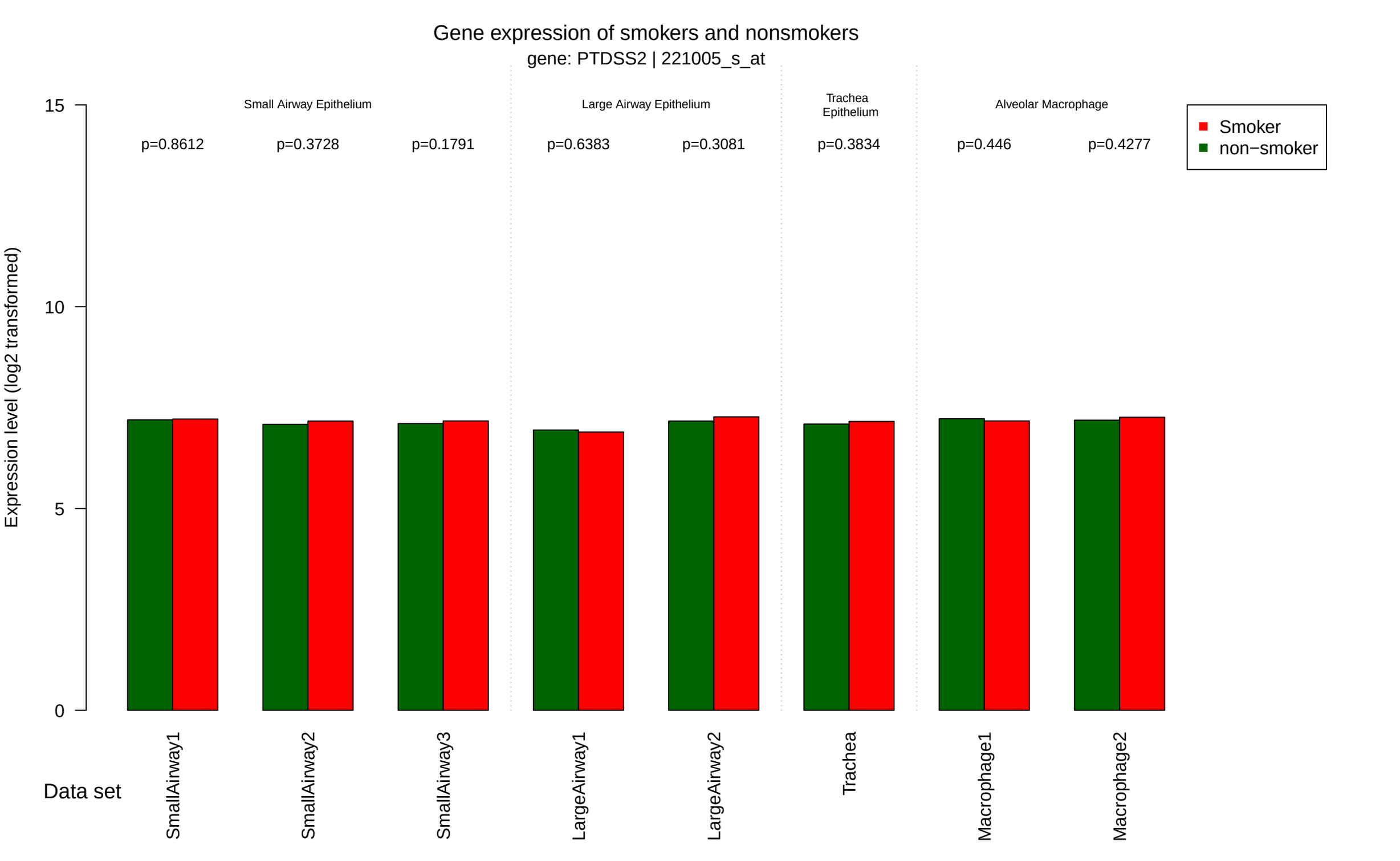
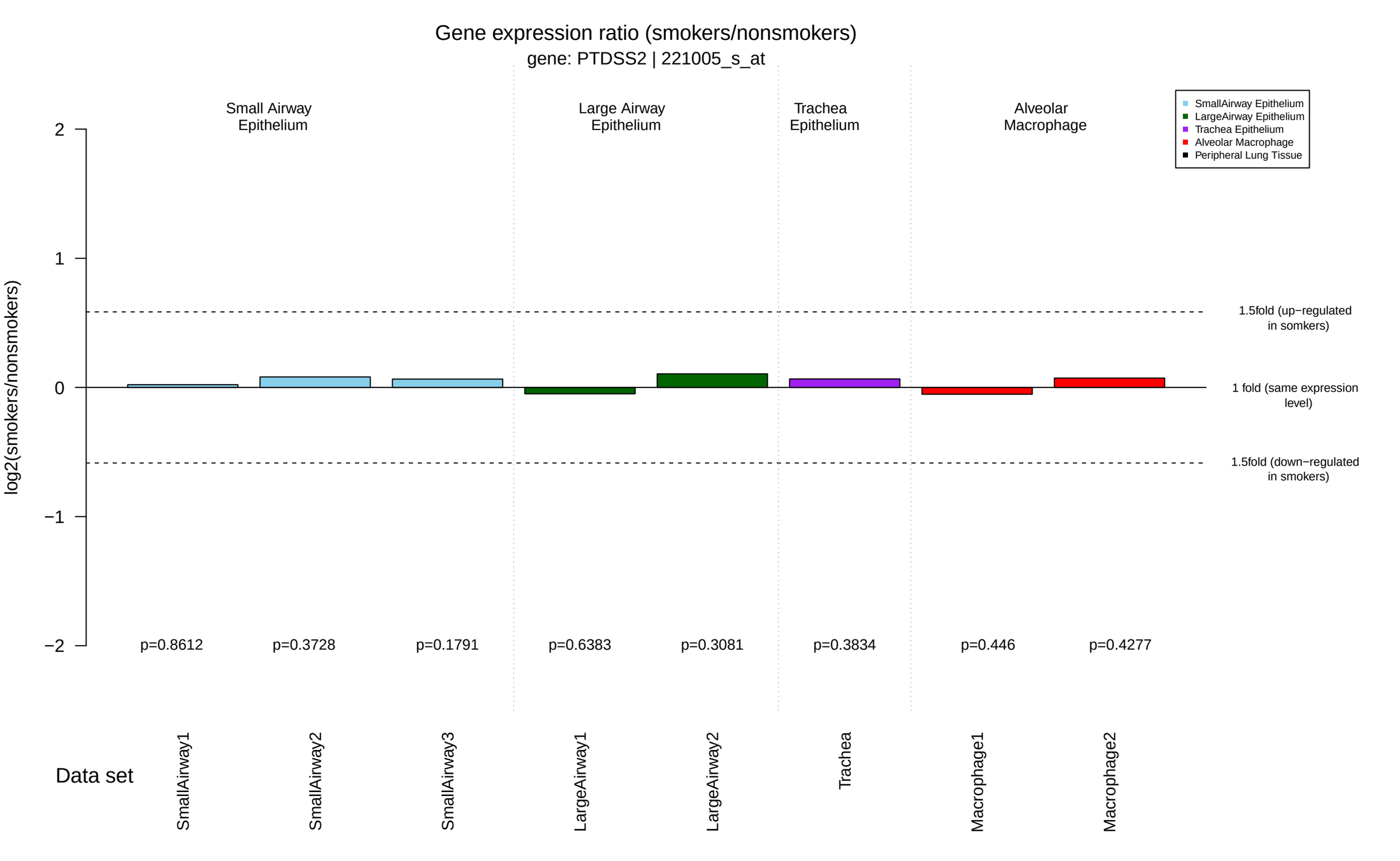
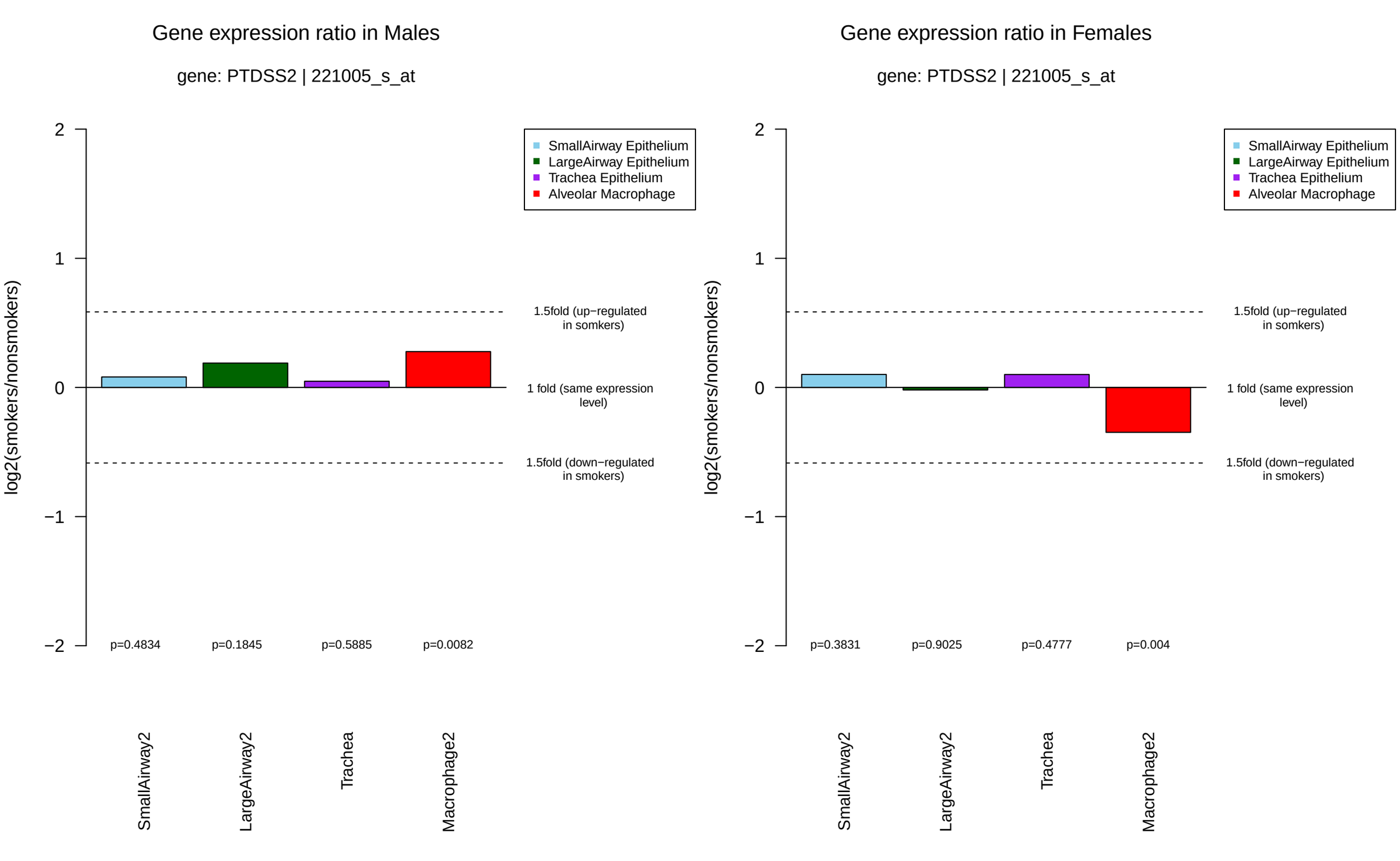
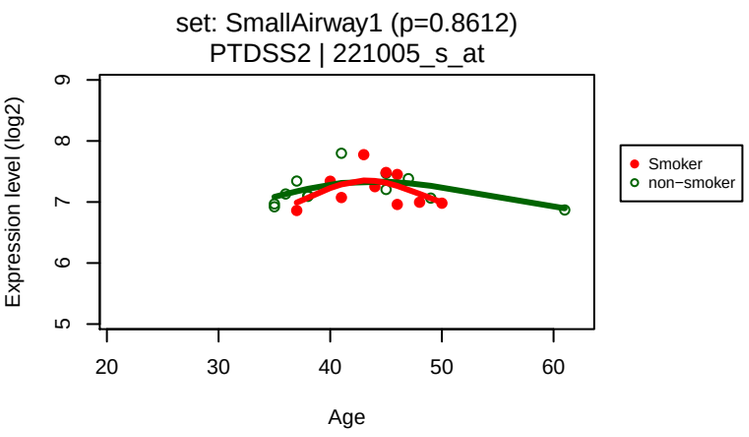
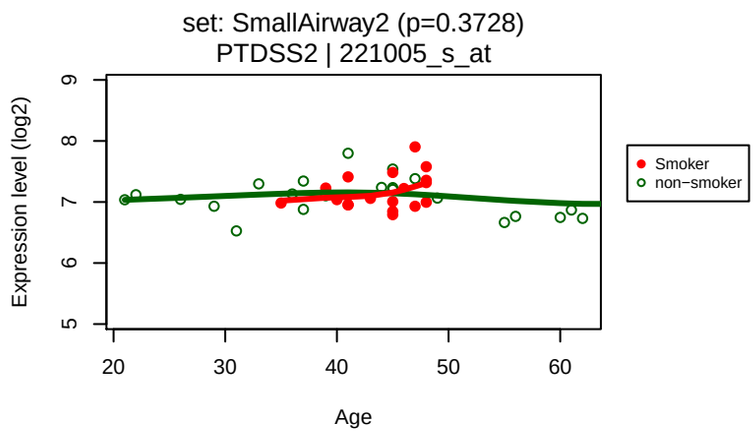
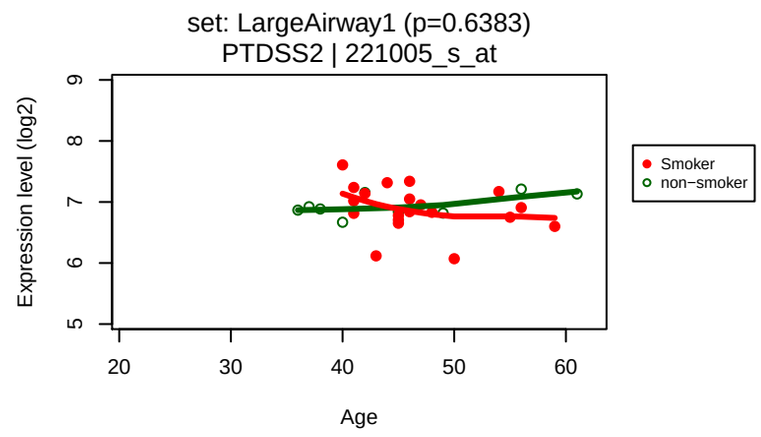

Gene expression of smokers and nonsmokers
gene: PTDSS2 | 221005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8612
p=0.3728
p=0.1791
p=0.6383
p=0.3081
p=0.3834
p=0.446
p=0.4277
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

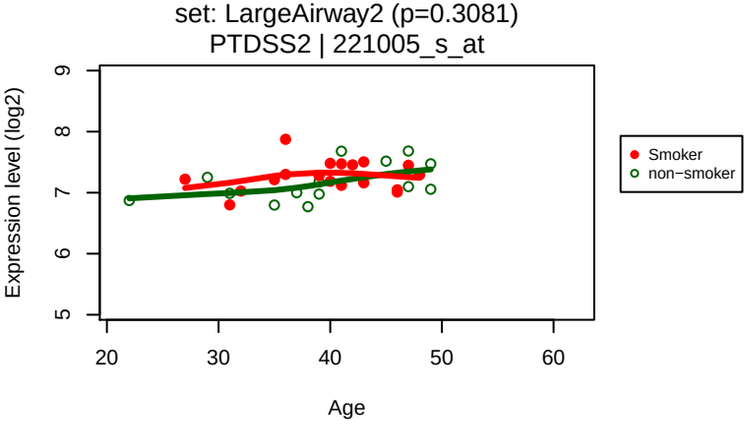

Gene expression ratio (smokers/nonsmokers)
gene: PTDSS2 | 221005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8612
p=0.3728
p=0.1791
p=0.6383
p=0.3081
p=0.3834
p=0.446
p=0.4277
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

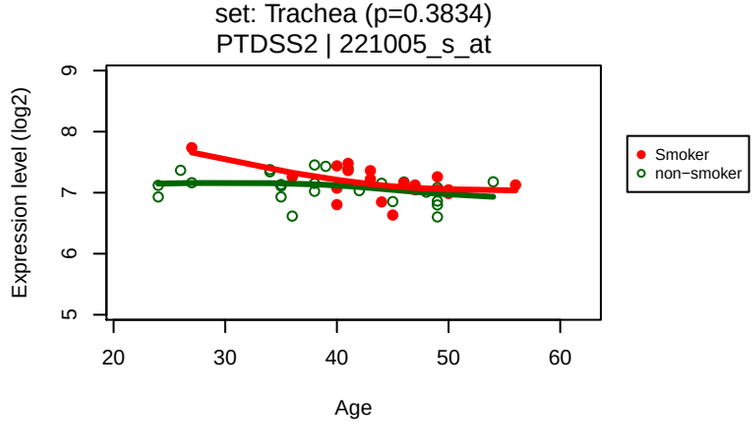

Gene expression ratio in Males
gene: PTDSS2 | 221005_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4834
p=0.1845
p=0.5885
p=0.0082
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTDSS2 | 221005_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3831
p=0.9025
p=0.4777
p=0.004
SmallAirway2
LargeAirway2
Trachea
Macrophage2

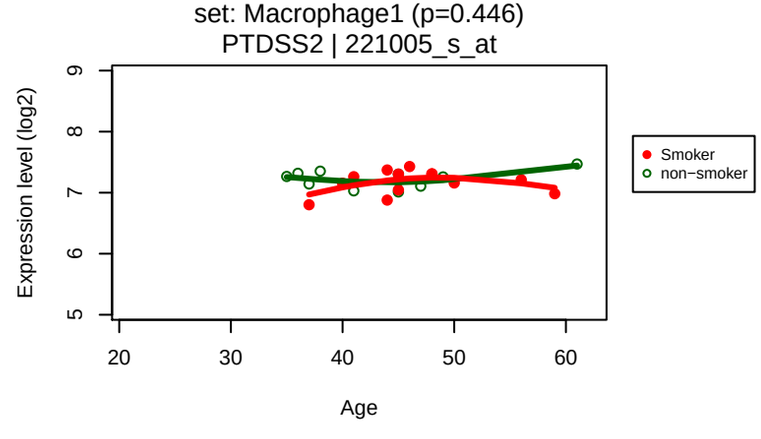

set: SmallAirway1 (p=0.8612)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker

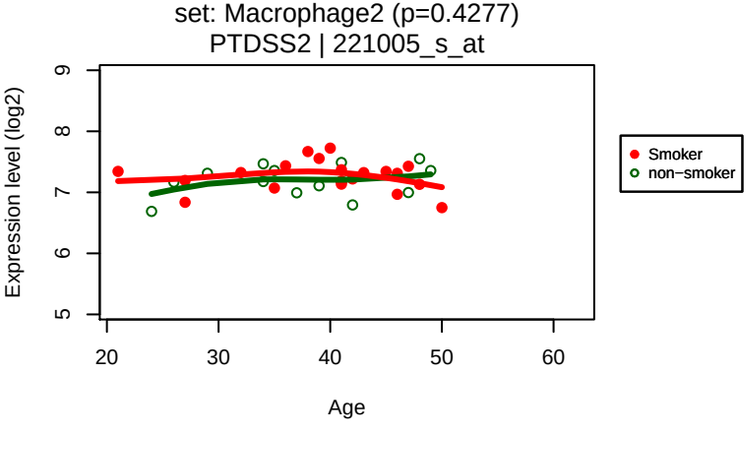

set: SmallAirway2 (p=0.3728)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker

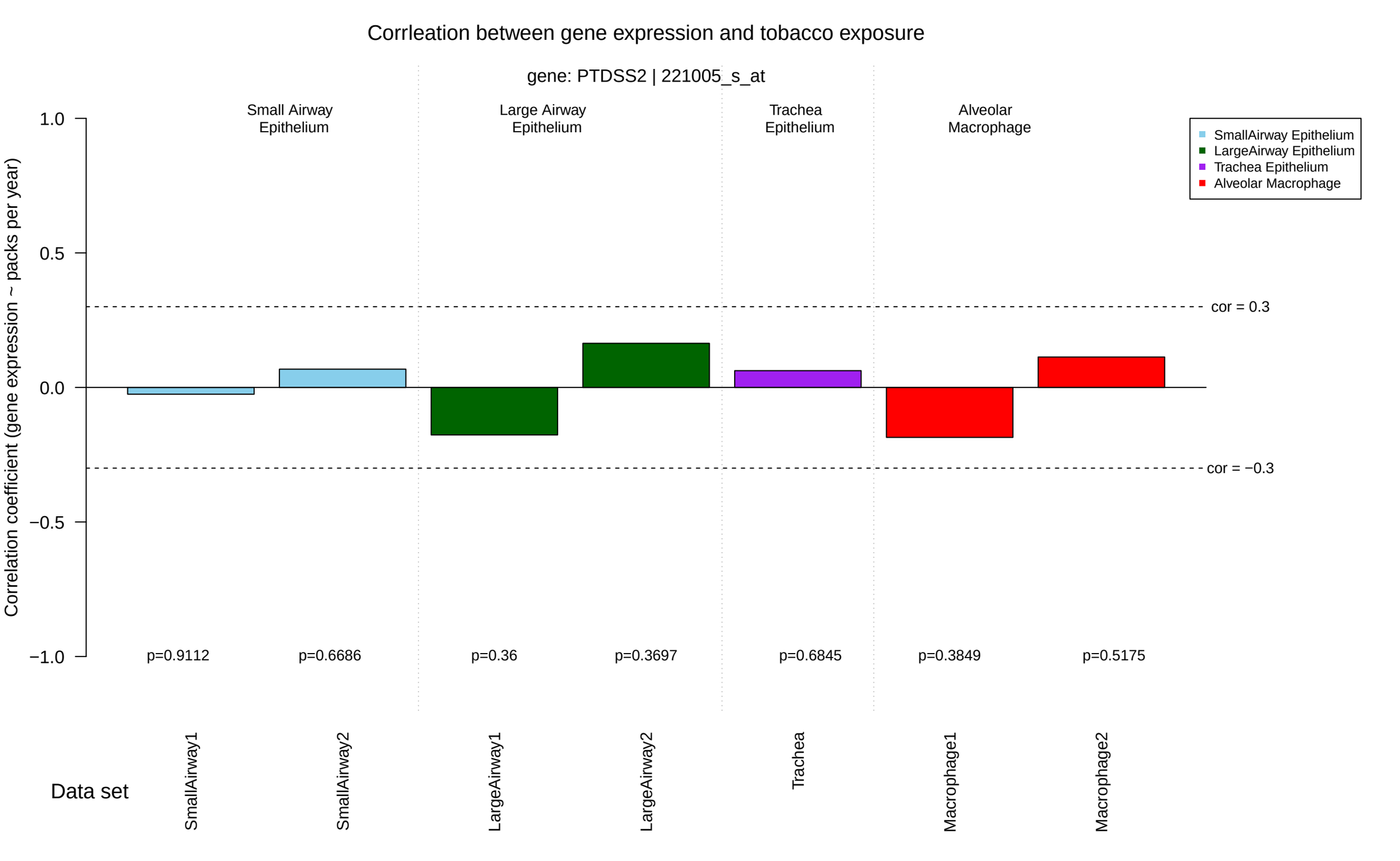

set: LargeAirway1 (p=0.6383)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3081)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3834)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.446)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4277)
PTDSS2 | 221005_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTDSS2 | 221005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9112
p=0.6686
p=0.36
p=0.3697
p=0.6845
p=0.3849
p=0.5175
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2